

DYN-EYE

YOLO Confidence Filter Evaluation Report

1. Objective & Context

The DYN-EYE pipeline relies on an initial YOLO v8 object detection filter to separate incoming inspection images into **known** defects (to be skipped) and **unknown** defects (to be routed to the VLM for discovery and annotation).

This evaluation report checks the robustness of this confidence thresholding logic. We tested the active YOLO model (trained on metal surface defects) using 44 novel plastic inspection images from the `data/filter_samples/` directory. Since the model has never seen plastic defects, the expected behavior is that the confidence scores should remain extremely low, causing the images to be correctly filtered as **unknown**.

2. Key Evaluation Metrics

The metrics from the active evaluation run are summarized below:

Metric	Value	Status / Interpretation
Total Images Evaluated	44	Full sample size of novel plastic images.
Classified as 'unknown' (Correct)	41 (93.2%)	Correctly filtered and successfully routed to VLM.
Classified as 'known' (Incorrect)	3 (6.8%)	Flagged due to over-confident bounding boxes.
Mean Confidence (All Detections)	0.0257	Extremely low, indicating high domain unfamiliarity.
Max Confidence Observed	0.4914	On image 'endface539.png' (Class: 'waist_folding').

3. Analysis & Recommendations

- **Active Threshold Robustness:** The current confidence threshold of **0.30** is highly effective, correctly identifying **93.2%** of the novel plastic images as unknown. This ensures that almost all novel domain data is successfully routed to the discovery loop.
- **False Positive Analysis:** 3 images (6.8%) were incorrectly flagged as 'known'. The maximum observed confidence was **0.4914** on image `endface539.png`, which mapped to the `waist_folding` defect class.
- **Calibration Recommendation:** If 100% correct filtering (0 false positives) is required for novel domains, the confidence threshold can be safely increased to **0.50**. Since the maximum hallucinated confidence was 0.4914, a 0.50 threshold completely eliminates incorrect classifications while maintaining high sensitivity for real known defects.

4. Dataset & File Breakdown

- **Total Files:** 44 PNG images inside `data/filter_samples/`.
- **Filename range:** `endface43.png` to `endface580.png`.
- **Over-confident files (>0.30 conf):** `endface539.png` (0.491), `endface556.png` (0.354), and `endface571.png` (0.312).